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**B.Tech. Degree V Semester Special Supplementary Examination
April 2022**

**ME 15-1504 THERMAL ENGINEERING
(2015 Scheme)**

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)



(10 × 2 = 20)

- I. (a) What is enthalpy of formation?
- (b) Write short notes on "excess air".
- (c) Write the difference between ideal and actual valve timing diagrams of a petrol engine.
- (d) How the Indicated power of a multi-cylinder engine is measured?
- (e) What is the effect of dissociation in spark ignition engines?
- (f) Write short notes on stratified charge engine.
- (g) State the various compressors used in supercharging the engines.
- (h) Explain the difference between Air cooling and Liquid cooling methods in I.C engines.
- (i) Write short notes on HUCR.
- (j) Write short notes on the various additives used in the diesel engines.

PART B

(4 × 10 = 40)

- II. A sample of fuel has the following percentage composition:
Carbon = 86 Percent, Hydrogen = 8 Percent, Sulphur = 3 Percent,
Oxygen = 2 percent, Ash = 1 Percent. For an air-fuel ratio of 12:1,
Calculate (i) Mixture strength as a percentage rich or weak
(ii) Volumetric analysis of the dry products of combustion.

OR

- III. One Kg of ethane (C_2H_6) is burned with 90% of theoretical air. Assuming complete combustion of hydrogen in the fuel determine the volumetric analysis of the dry products of combustion.
- IV. A Diesel engine working on a dual combustion cycle has a stroke volume of 0.0085 m^3 and a compression ratio of 15:1. The fuel has a calorific value of 43890 KJ per Kg. At the end of the suction, the air is at 1 bar and 100°C . The maximum pressure in the cycle is 65 bar and air fuel ratio is 21:1. Find for ideal cycle the thermal efficiency. Assume $C_p = 1$ and $C_v = 0.71$.

OR

(P.T.O.)

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- V. A 4-cylinder petrol engine has a bore of 60 mm and a stroke of 90 mm. Its rated speed is 2800 r.p.m. and is tested at this speed against brake which has a torque arm of 0.37 m. The net brake load is 160 N. and the fuel consumption is 8.986 litres per hour. The specific gravity of the petrol used is 0.74 and it has a lower calorific value of 44100 KJ per Kg. A Morse test is carried out and the cylinders are cut out in the order 1, 2, 3, 4 with corresponding brake loads of 110, 107, 104 and 110 N respectively. Calculate for this speed (i) Engine Torque (ii) Brake mean effective pressure (iii) Brake thermal efficiency (iv) Mechanical efficiency (v) Indicated mean effective pressure.
- VI. Explain the following methods of governing I.C engines with a neat sketch (i) Hit and miss method (ii) Quality governing (iv) Quantity governing.
- OR**
- VII. Explain the following with a neat sketch
(i) Free piston Engine (ii) Wankel Engine.
- VIII. Explain the phenomenon of auto-ignition. Explain how auto-ignition is responsible for Knocking in S.I engines
- OR**
- IX. With a neat sketch explain the various combustion chambers used in C.I engines.
